

Sets & Set Theory

Here are some problems for class discussion. Explain your reasoning on every problem.

The notation $P(A)$ represents the power set of set A , which is the set of all subsets of A . That is, $P(A) = \{X \mid X \subseteq A\}$.

(1) Let $A = \{a, b, c, d, 2, 4, 6\}$. Answer true or false to the following statements. No explanations are needed.

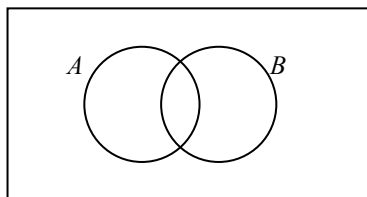
- (a) _____ $b \in A$
- (b) _____ $7 \in A$
- (c) _____ $\{6\} \in A$
- (d) _____ $\emptyset \subseteq A$
- (e) _____ $\{2, 4, 6, 8\} \subseteq A$
- (f) _____ $A \in \{d\}$
- (g) _____ $\{a, b, c, d\} \subseteq A$
- (h) _____ $\{a, b, c, d\} \in P(A)$
- (i) _____ $\{\emptyset\} \in P(A)$
- (j) _____ $\emptyset \subseteq P(A)$

(2) True or false? No explanations are needed.

- (a) _____ $3 \in \mathbb{Z}$
- (b) _____ $\{3\} \in P(\mathbb{Z})$
- (c) _____ $\emptyset \in P(\{\emptyset, \{\emptyset\}\})$
- (d) _____ $\emptyset \subseteq (A \cap \bar{B}) - (C \cup (D - A))$, where A, B, C , and D are sets.

(3) Let $B = \{b, \{a, b\}\}$. Find $P(B)$.

(4) Shade the set $(A \cap B) \cup (A - B)$ in the Venn diagram given below:



(5) Let $A = \{1, 3, 6, 9\}$, $B = \{1, 2, 4, 6, 8\}$, and $C = \{1, 3, 5, 7, 9\}$

(a) Find $C - (B - A)$.

(b) Find $(C - B) - A$.

(6) Recall that the power set $P(A)$ of a set A is the set of all subsets of A . Suppose that $A = \{1, 2, 3, 4, 5\}$.

(a) List three elements of $P(A)$.

(b) How many elements are there in $P(A)$? Explain with a counting argument (not citation of a theorem).

(c) List three elements of $A \times P(A)$.

(d) How many elements are there in $A \times P(A)$? Explain with a counting argument.

(e) How many elements are there in $P(A \times P(A)) \cap A$? Explain.

Proving results involving sets.

(7) Prove or disprove: If $A \subseteq \bar{B}$ then $B \subseteq \bar{A}$.

(8) Let A , B , C , and D be sets with $C \subseteq A$ and $D \subseteq B$. Prove or disprove: $D - A \subseteq B - C$.

(9) Let A , B , and C be sets. Prove or disprove: If $A \cap C \subseteq B \cap C$ then $A \subseteq B$.